

'O' notation value in a number-crunching zone, which will have a big impact. If you are working on products that are soft real-time or hard real-time, it is very important that you have an algorithm expert with you, to nip any algorithmic issues in the bud.

- If you do not have the resources to accept the transaction, then reject it and return a negative acknowledgement immediately. Do not accept the transaction and later find out that the system could not process it.
- Critical sections must be synchronised with proper synchronisation mechanisms, else it will affect system performance.
- Resources that are not released properly, such as memory, span, chassis, threads or processes, will affect system performance over a period of time.
- Use of fancy programming languages (because the developer or the team feel it is nice to use, or are more confident in using a certain language and a more appropriate skill-set is not available) often lowers system performance.

Generally, in soft real-time and hard real-time systems, we go for assembly, C or C++ or any other language in which the performance considerations are sufficiently met, and the compilers have matured.

Summary

In this article, we looked at how to quantify the good quality of an architecture. We also discussed pragmatic ways to improve software, and hence performance. We looked at eight different kinds of software performance parameters. However, there are many more software performance parameters that are dependent on the product. For example, a television has a performance parameter of how quickly a channel change is reflected on the screen. LCD monitors have their screen resolution as one of the parameters. Starting with the next article, we will discuss in detail the different subsystems in the Linux kernel, such as file systems, IPC, input/output and memory management. We will elaborate on what design considerations an architect has to be aware of while creating products on the Linux platform. 

References

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